Coding & Solution

Date	15 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/sushma-boya/A-Comprehensive-Measure-of-Well-Being
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Key Features Implemented	• Human Development Index (HDI) prediction using Machine Learning. • Data preprocessing and cleaning. • Linear Regression model training and evaluation. • Responsive Flask web application. • Prediction of HDI score and development category (Low, Medium, High, Very High). • User-friendly interface with real-time prediction results.
Pending / Incomplete Features	• User authentication. • Database integration. • Interactive data visualization dashboards. • Cloud deployment for online access.
Setup / Run Instructions	1. Clone the GitHub repository. 2. Install dependencies using `pip install -r requirements.txt`. 3. Run `python app.py`. 4. Open `http://127.0.0.1:5000` in a web browser. 5. Enter the required input values and click **Predict** to view the HDI prediction.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / modules	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes *(basic input validation and exception handling)*
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture by separating the machine learning model, Flask application, templates, static resources, and dataset into organized directories. The application successfully predicts the Human Development Index (HDI) using socio-economic indicators and demonstrates the integration of Machine Learning with web application development. The GitHub repository includes the complete source code, documentation, screenshots, and setup instructions, making the project easy to understand, reproduce, and extend in the future.